

Tseng, Chi-Teng, MD/MPH
Normal MSK in Children

Normal Anatomy of MSK in children

Knee Joint

理論

實作

講師簡介

基隆長庚小兒科 曾吉騰醫師

異位性皮膚炎 | 過敏性鼻炎 | 氣喘 | 關節超音波 | 兒童風濕疾病

學經歷

中國醫藥大學公共衛生碩士
中國醫藥大學醫學系
國立成功大學醫院 兒科部住院醫師
嘉安婦幼診所小兒科醫師
林口長庚兒童氣喘過敏風濕科研究醫師
歐洲兒童風濕病學會關節超音波進修

專科證照

中華民國小兒科專科醫師
中華民國免疫學會免疫過敏專科醫師
台灣兒童過敏氣喘免疫及風濕病醫學會兒童過敏免疫專科醫師
台灣兒科醫學會青少年專科醫師
台灣兒科醫學會兒童發展與健康篩檢服務醫師
台灣胸腔暨重症加護醫學會健保氣喘慢性照護醫師
中華民國超音波學會會員

個人部落格



Certificate of Attendance

We hereby certify that

CHI TENG TSENG

Has actively and successfully participated in the PReS musculoskeletal ultrasound course - Intermediate level held in Rotterdam, the Netherlands from 25 to 27 September 2023.

The course was organized by the PReS Imaging Working Party and was endorsed by EULAR.

Rotterdam, Netherlands, 27 September 2023

Linda Rossi

France

Secretary of Imaging Working Party

Daniel Windschall

Germany

Chair of Imaging Working Party

Learning objectives

- Technique of Performing Ultrasound
- Common Normal MSK Ultrasound Findings
- Hand-on time

Tips

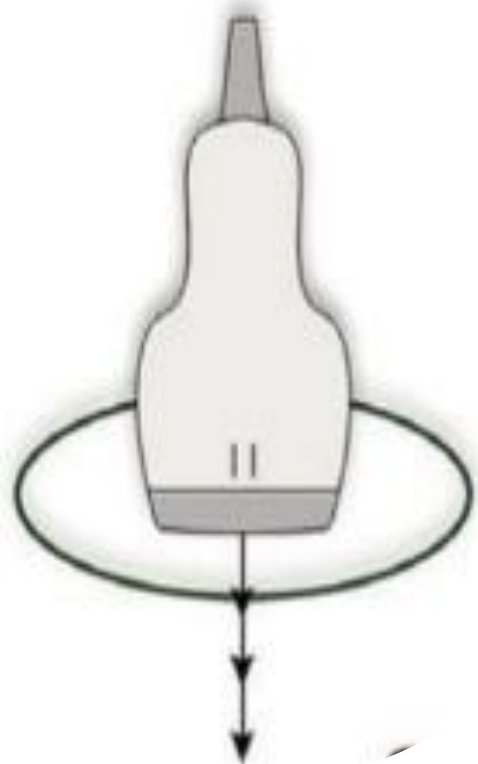
握好探頭



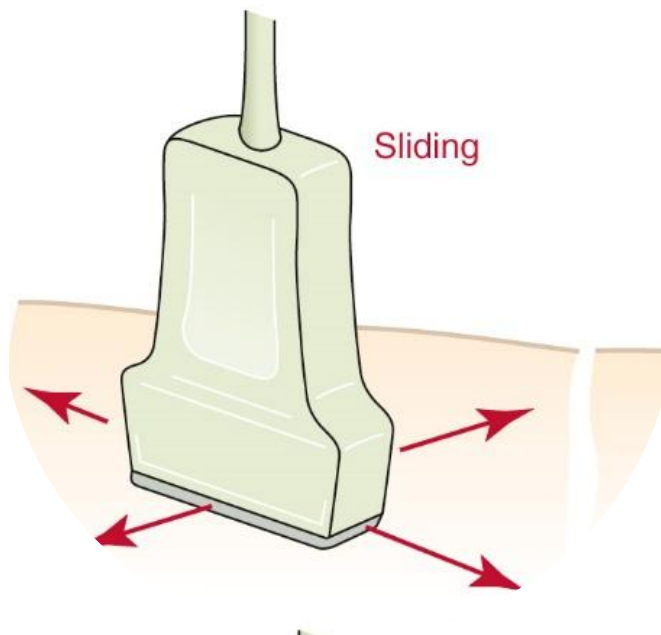
Tips

探頭操作

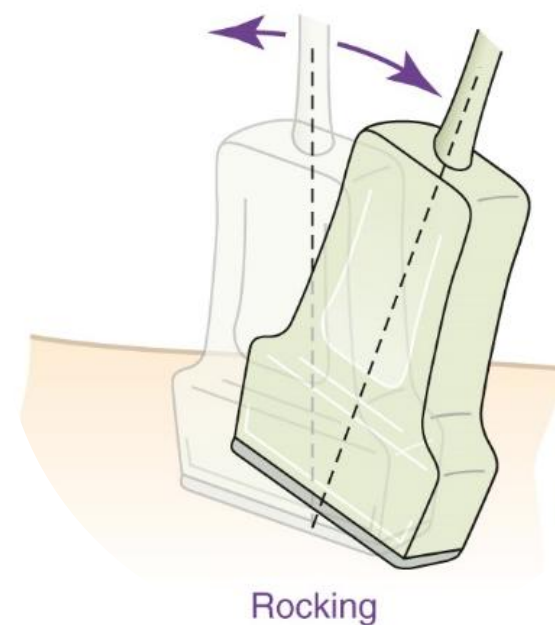
壓



滑



搖



Probe Placement **Mark**

向上

向前

向內

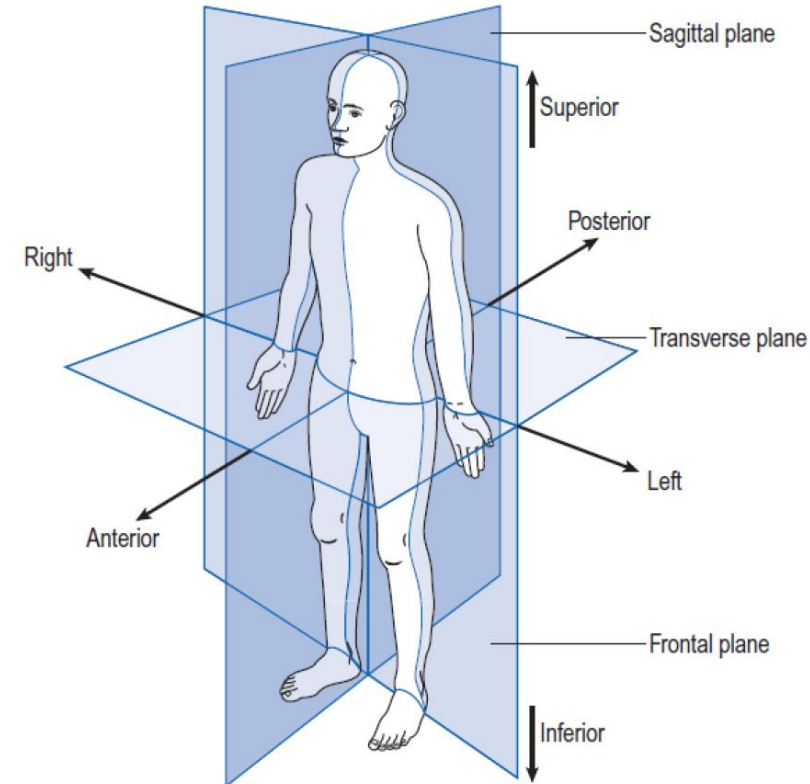


Figure 1. The anatomical position, with three reference planes and six fundamental directions
M. Whittle, Gait analysis : an introduction. Butterworth-Heinemann, 2007.

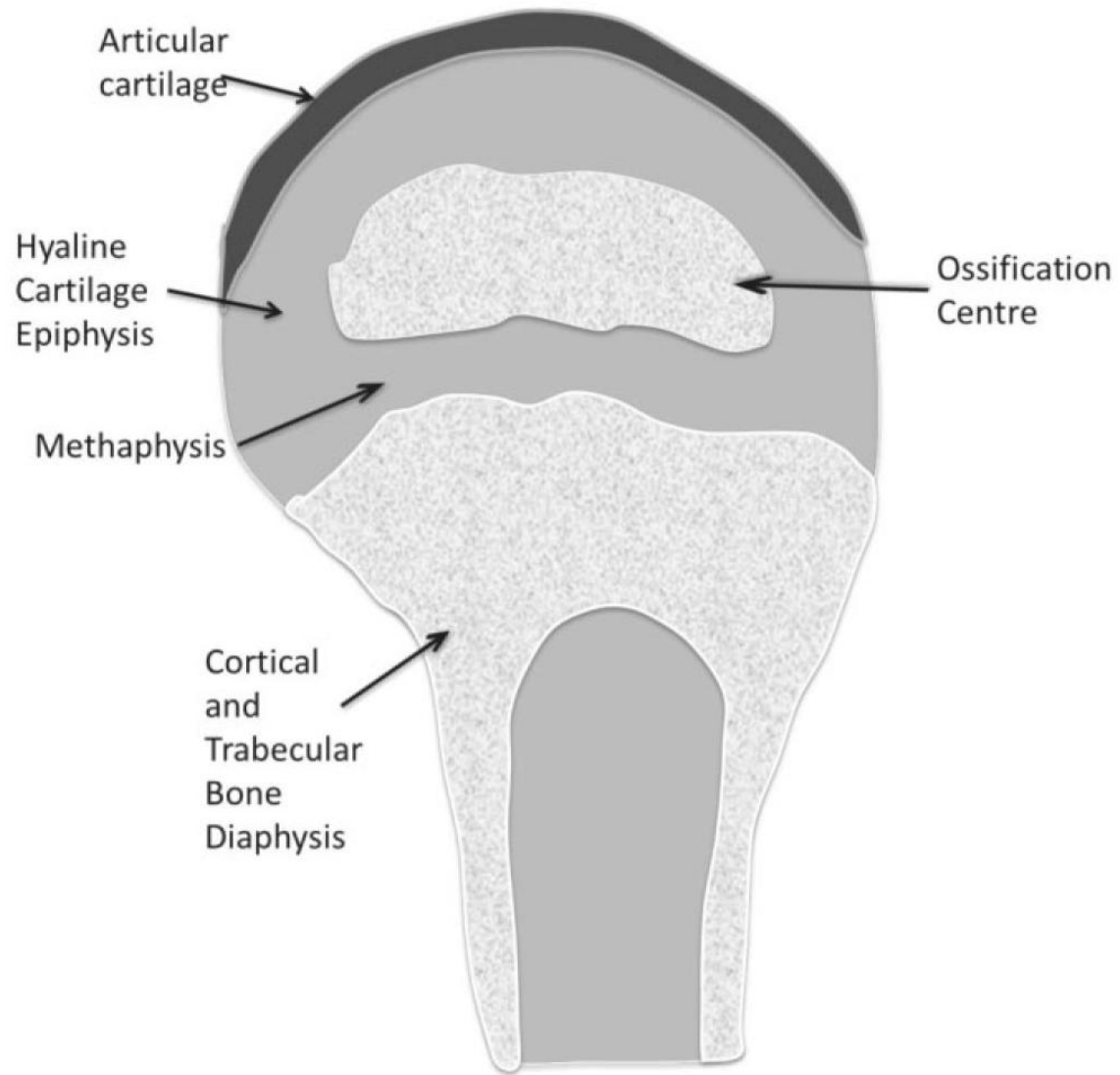
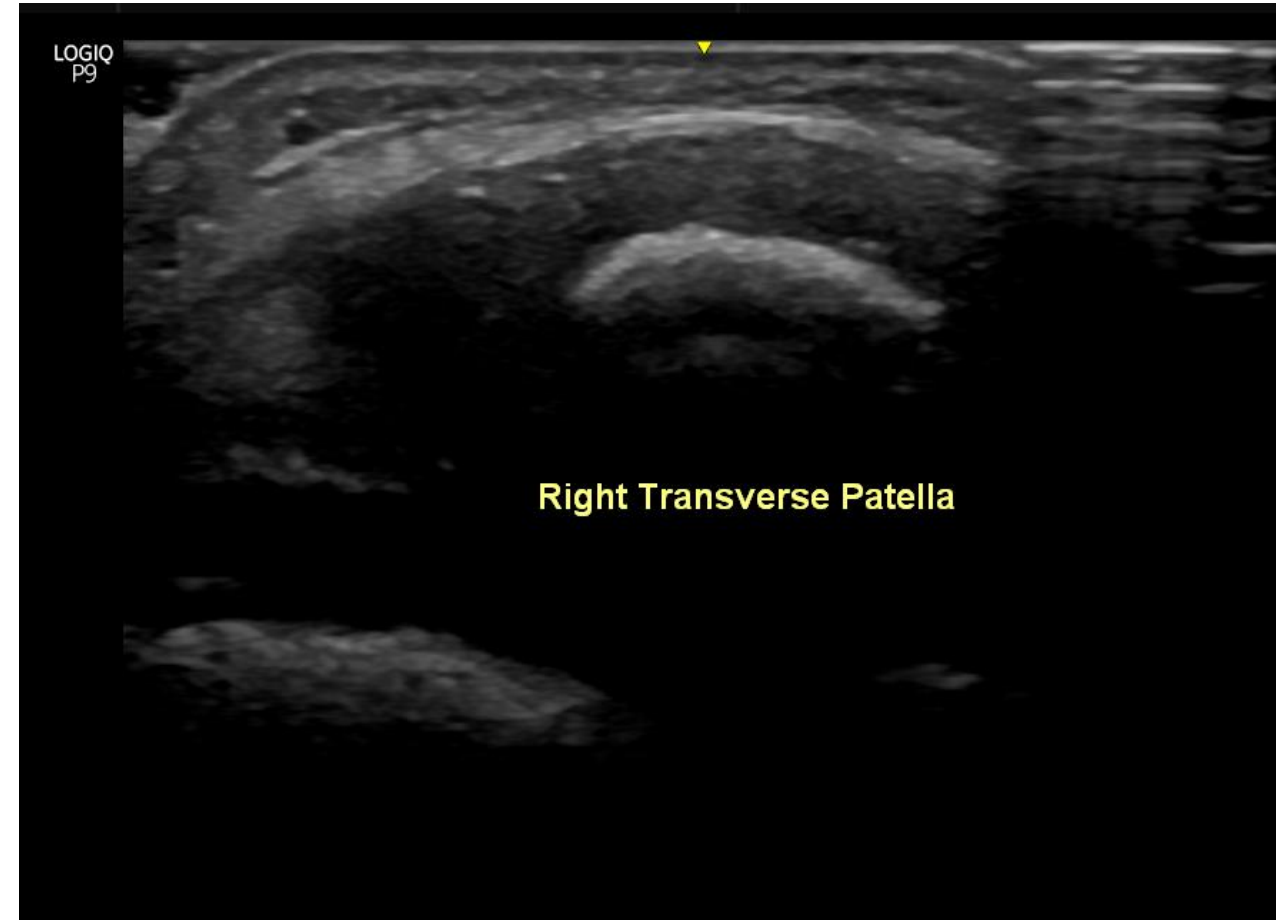


Figure 1. Schematic outline of the various structures within a bone forming a joint.



5-year-old boy

Arthritis Care Res (Hoboken). 2015 Jan;67(1):136-42

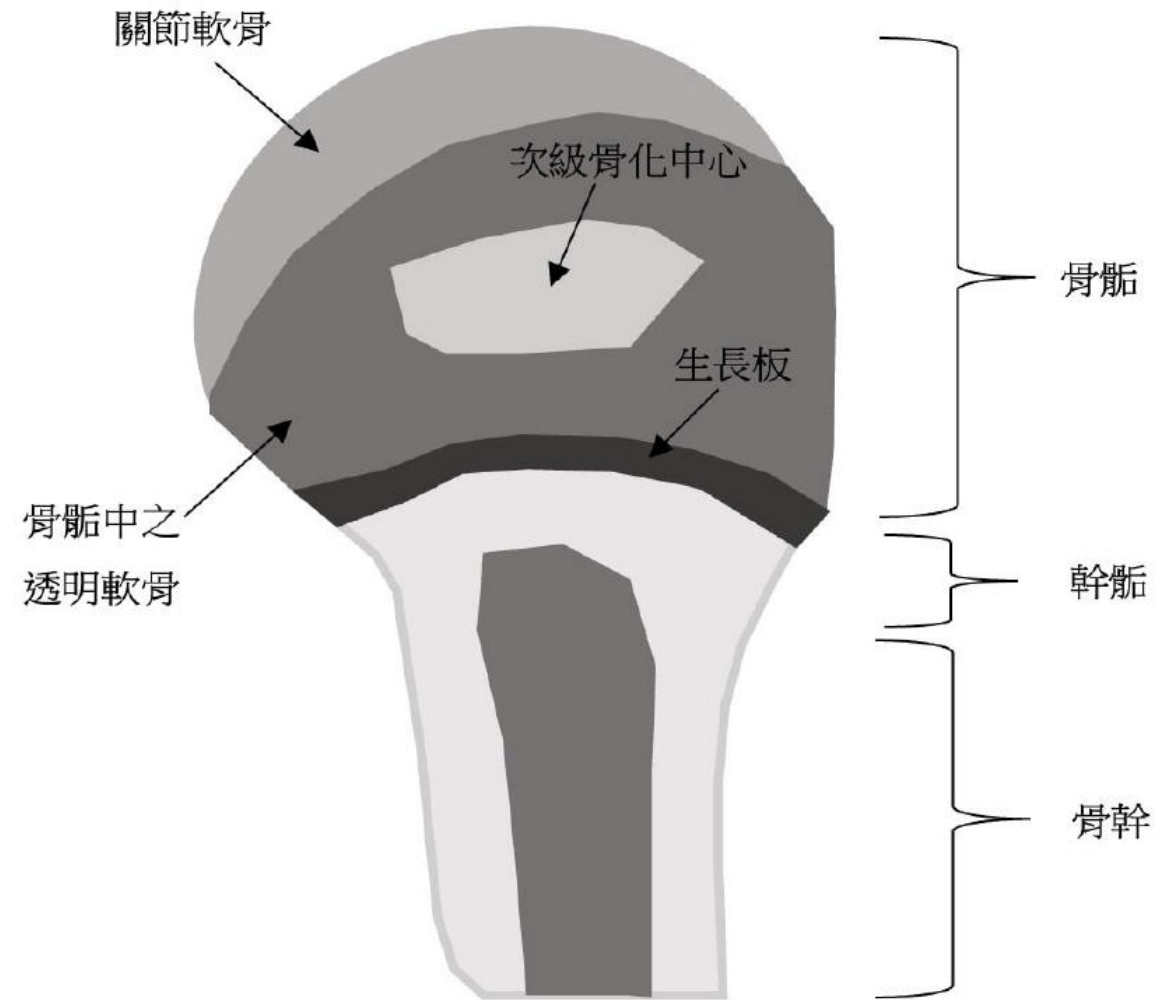


圖 1. 未成熟骨頭的結構示意圖(以長骨為例)

Definition 1

The hyaline cartilage will present as a well-defined anechoic structure (with/without bright echoes/dots) that is non-compressible. The cartilage surface can (but does not have to) be detected as a hyperechoic line.

Definition 2

With advancing maturity, the epiphyseal secondary ossification center will appear as a hyperechoic structure, with a smooth or irregular surface within the cartilage.

Definition 3

Normal Joint Capsule - A hyperechoic structure which can (but does not have to) be seen over bone, cartilage and other intraarticular tissue of the joint.

Definition 4

Normal synovial membrane - Under normal circumstances, the thin synovial membrane is undetectable.

Definition 5

The ossified portion of articular bone is detected as a hyperechoic line. Interruptions of this hyperechoic line may be detected at the growth plate and at the junction of two or more ossification centers.

Figure 6. The final versions of the definitions.

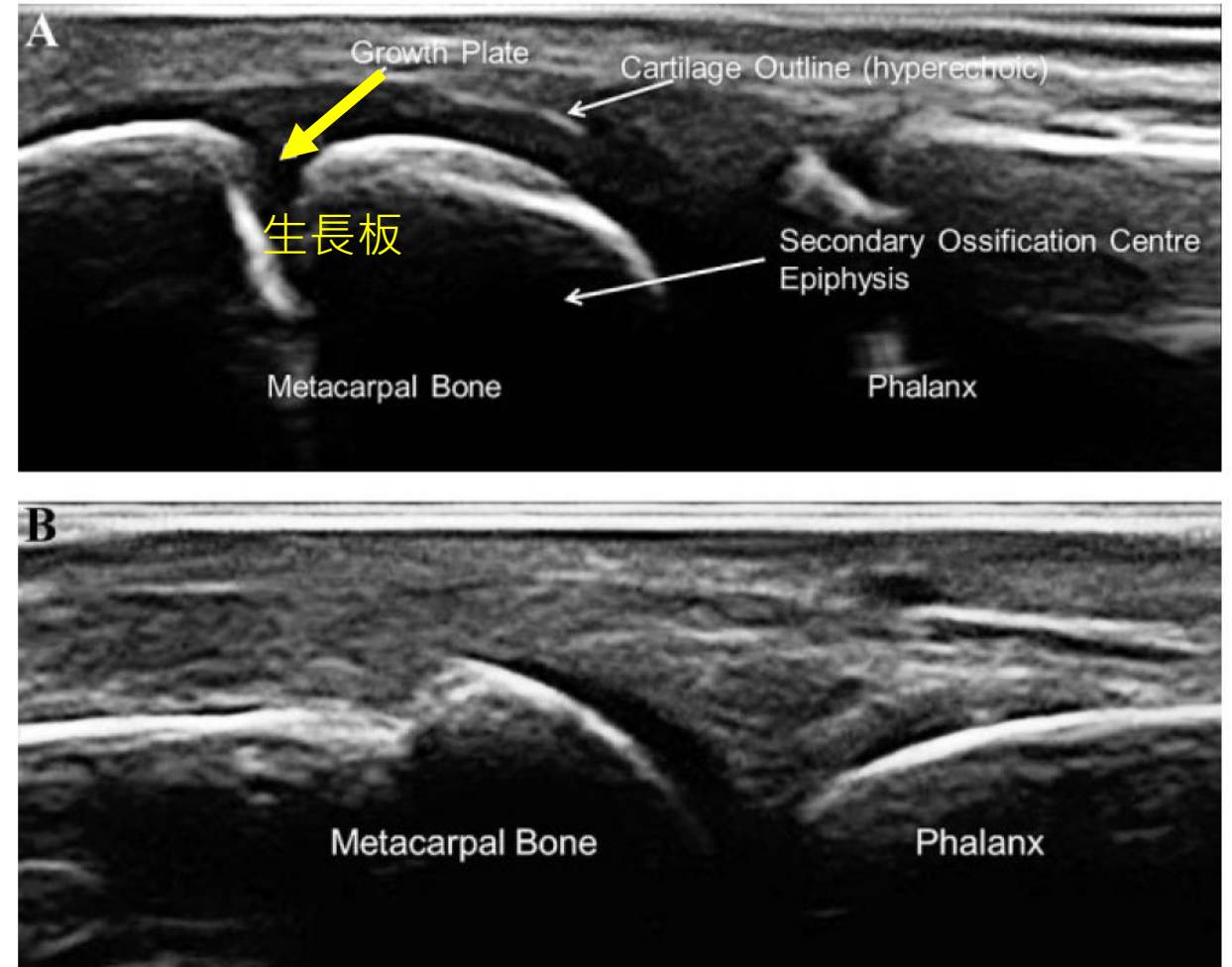
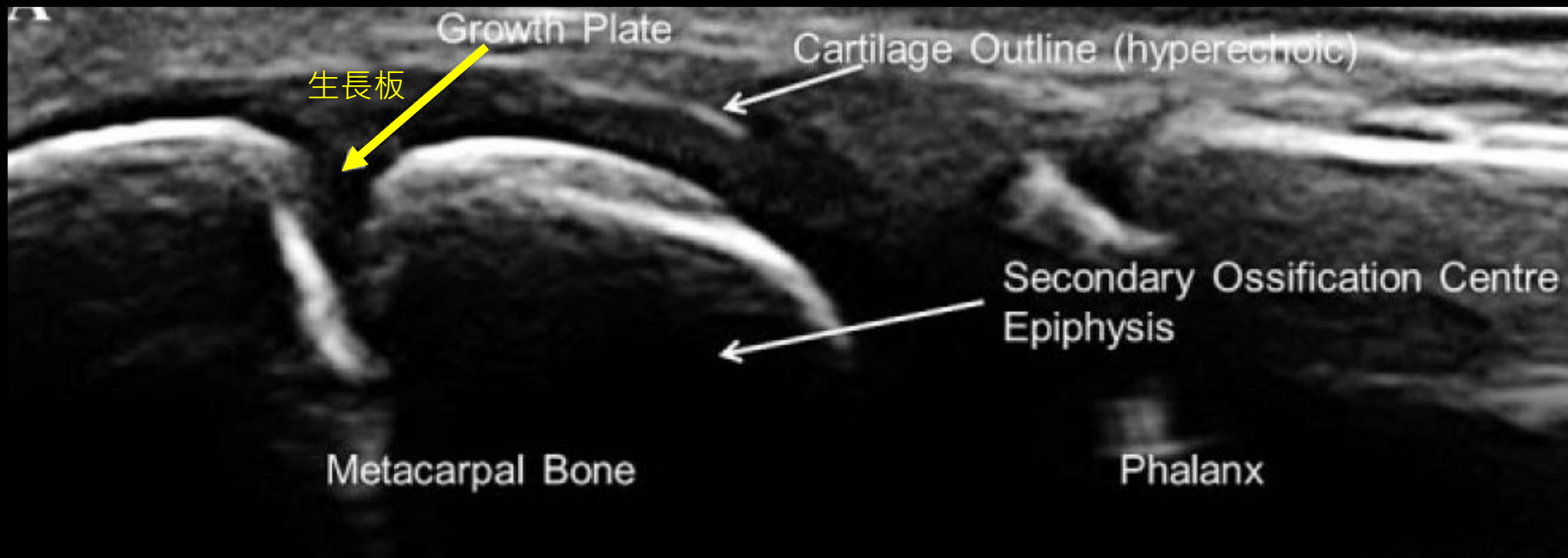


Figure 2. A, Ultrasonographic image of a metacarpophalangeal (MCP) joint in longitudinal view of a 10-year-old. This gray-scale

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成長中骨骼超音波表現

發育未全的骨骼解剖結構

- **骨骺 (Epiphysis)**：主要由透明軟骨組成（無回音 [anechoic]）。
- **生長板 (Physis)**：於幹骺端與骨骺之間呈現為無回音間隙。
- **骨化中心 (Ossification Nuclei)**：出現年齡不一；具有強回音邊緣及後方聲影。

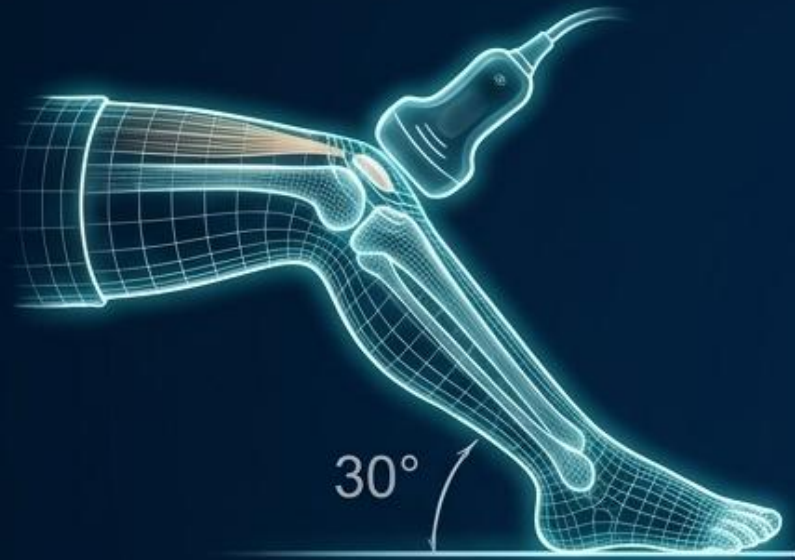
關鍵判別要點

- 軟骨厚度隨年齡增長而**顯著減少**。
- 男性軟骨厚度通常較女性為**厚**。
- 生長板不規則屬**正常現象**（並非侵蝕或骨折）。

Navigating the Suprapatellar Recess

Standard 30° Flexion Protocol

Longitudinal View



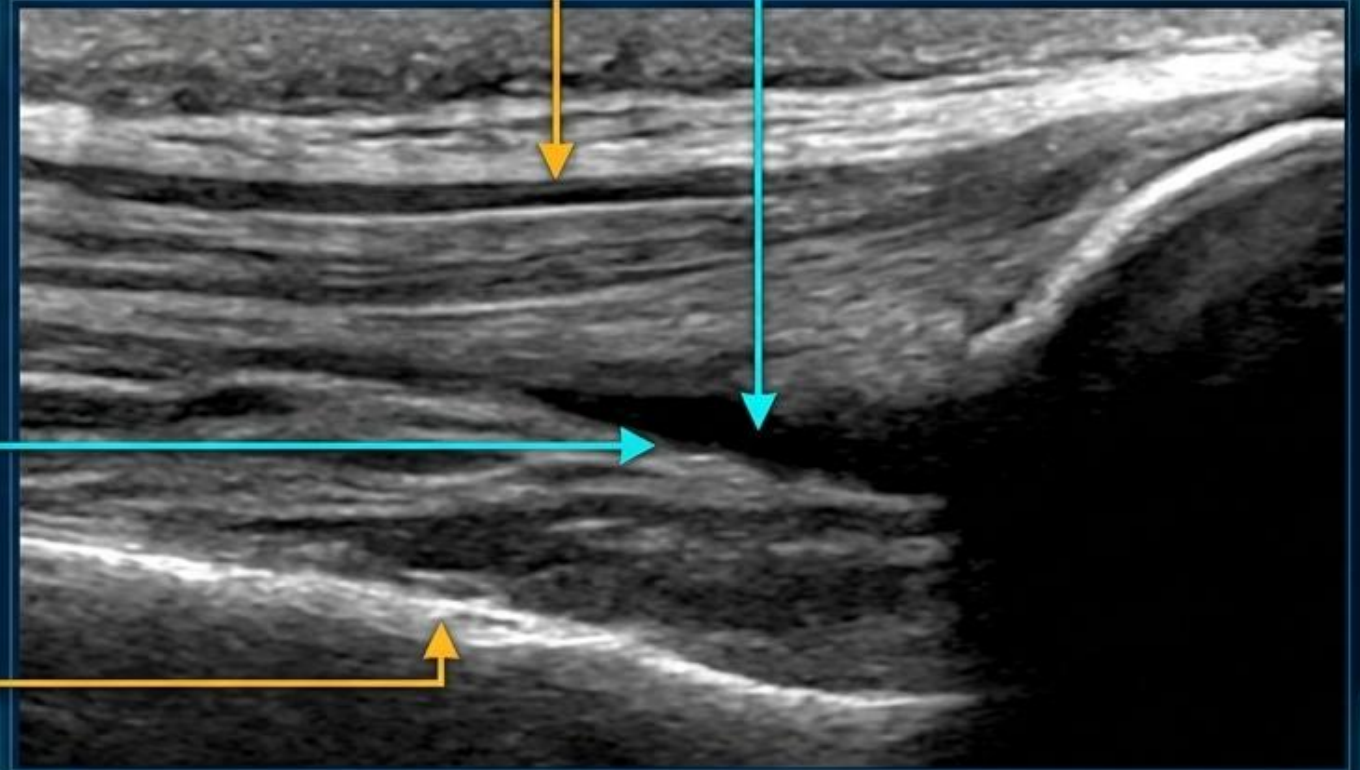
Quadriceps Tendon
(Fibrillar pattern)

Suprapatellar Recess
(Target zone for fluid)

Thick Anechoic Patellar Cartilage

Warning: Dark void—Do not mistake for fluid!

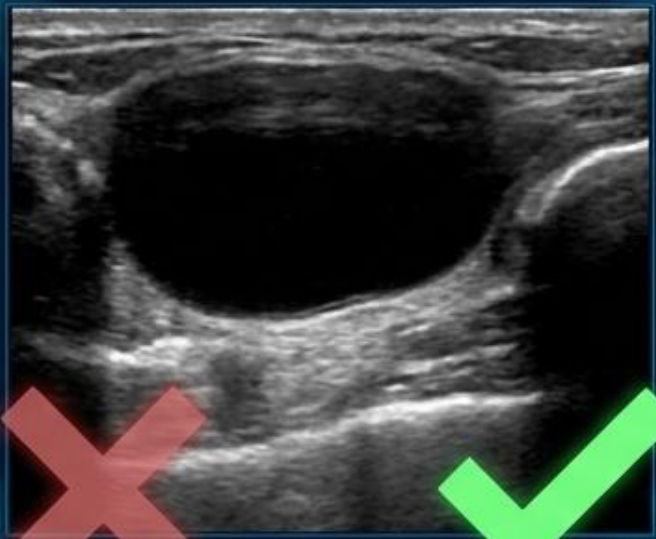
Femur
(Hyperechoic bony baseline)



Three 'False Alarms' in the Pediatric Knee

Common Diagnostic Pitfalls to Avoid Today

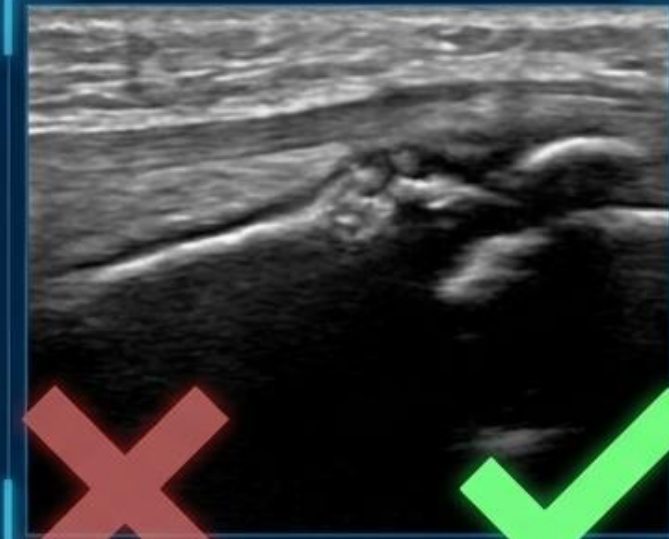
1. The Void



Massive
Effusion

Normal Thick
Anechoic
Cartilage

2. The Break



Avulsion Fracture /
Osgood-Schlatter

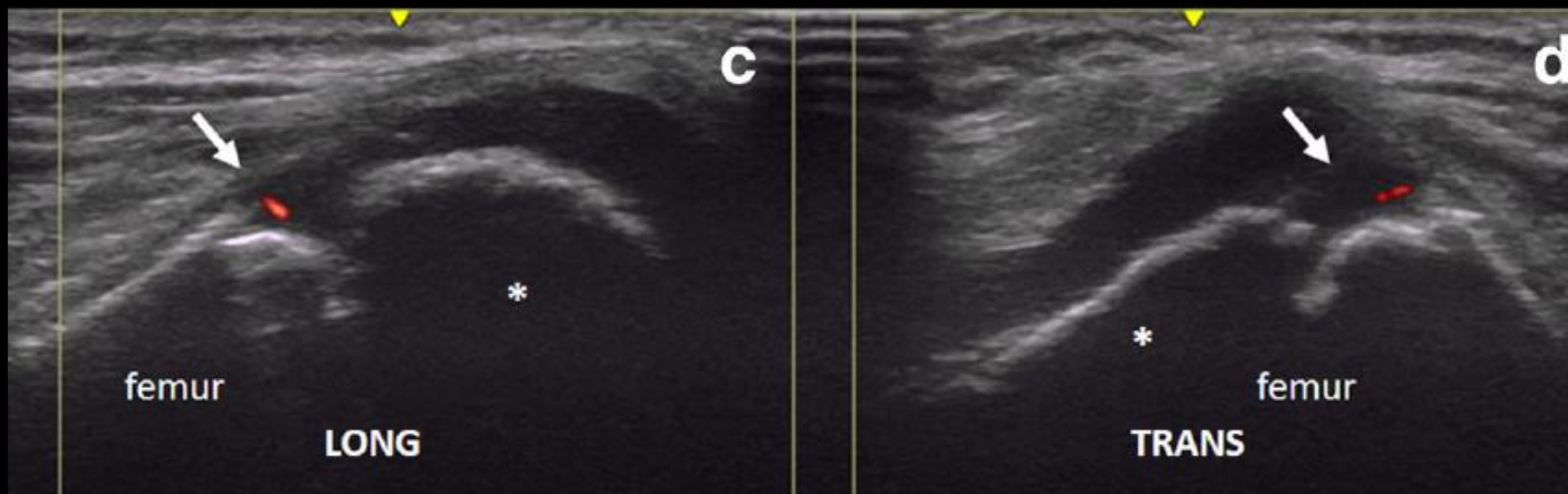
Normal
Secondary
Ossification
Center

3. The Fire

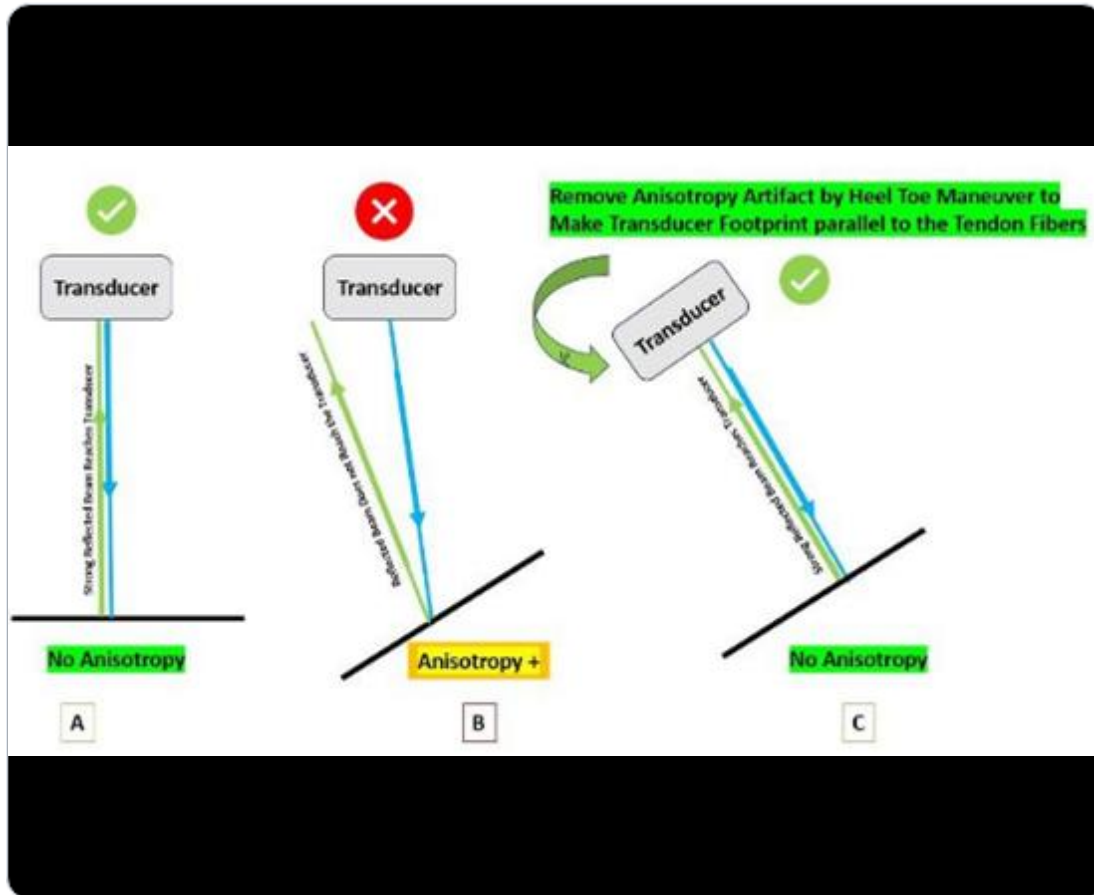


Hyperemic
Inflammation

Normal Vascular
Channels in
Growing Cartilage



Avoiding Pitfalls: Beginner Tips



1. Manage Anisotropy

Tendon fibers must be **90° perpendicular**. Angling away makes healthy tendons look "dark" (anechoic), mimicking tears.

2. Light Touch

Children's joints are small. Excessive probe pressure can **collapse small effusions** and squeeze out Doppler signals.

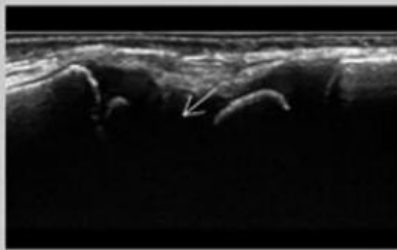
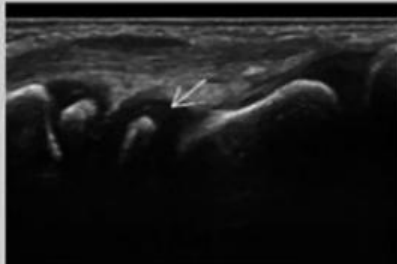
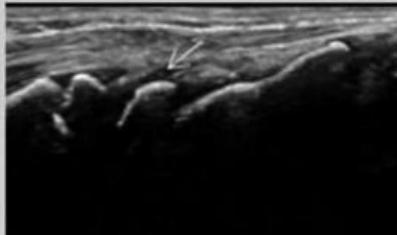
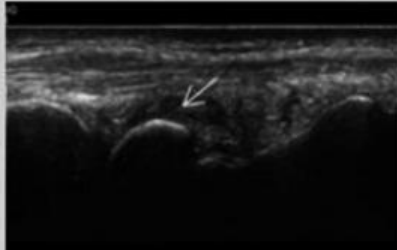
3. PRF Settings

Use "Low Flow" presets (PRF ~ 0.6 - 0.8 MHz) to detect low-velocity physiological and inflammatory flow.

Diagnostic Benchmarks: Knee Joint

Feature	Healthy Findings	Abnormal (Pathological)
Joint Fluid (Recess)	< 3mm is common. Often contains "speckles".	Significant distension (> 3-4mm) or bulging.
Synovial Layer	Undetectable or thin double-layer (Mdn: 1.5-2mm).	Thickened, hypoechoic, non-displaceable.
Doppler Signal	Located in cartilage canals or fat pads.	Located within the synovial hypertrophy.
Ossification	Irregular surfaces in maturing skeletons.	Frank cortical breakage (Erosion).

The steps involved in the skeletal development of long bones, small bones & patella on a 4- degree scale

Ossification grade 0	Non-ossified epiphyseal bone, short bones or patella.	
Ossification grade 1	Small ossification centers, dominant cartilage, visible growth plate.	
Ossification grade 2	Large ossification centers, thin cartilage, visible growth plate.	
Ossification grade 3	Complete ossification.	

*With Permission from
Dr. Shu-Mei Yang*

Take Home Message

- 壓、滑、搖
- 軟骨、生長板、骨化中心
- Echogenecity; Compressible; PD + Color Doppler

更多教學內容可參考

↓↓↓部落格↓↓↓



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吳昭儀 教務長



Dr. Clara Malattia

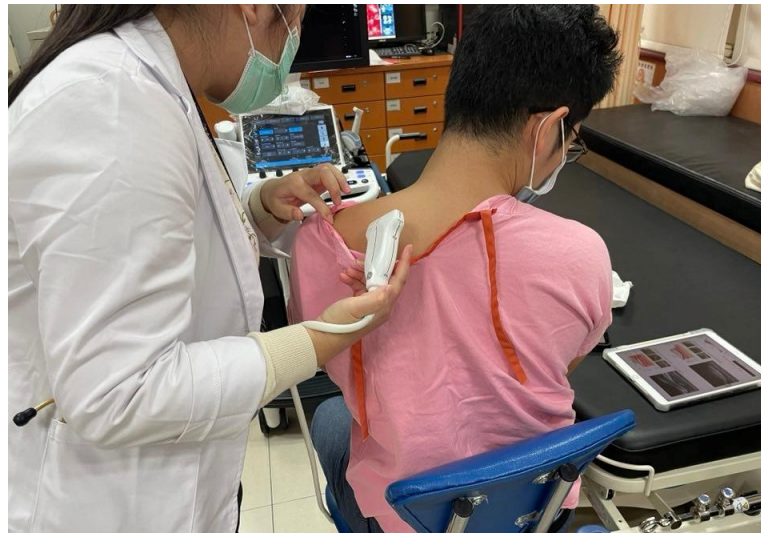
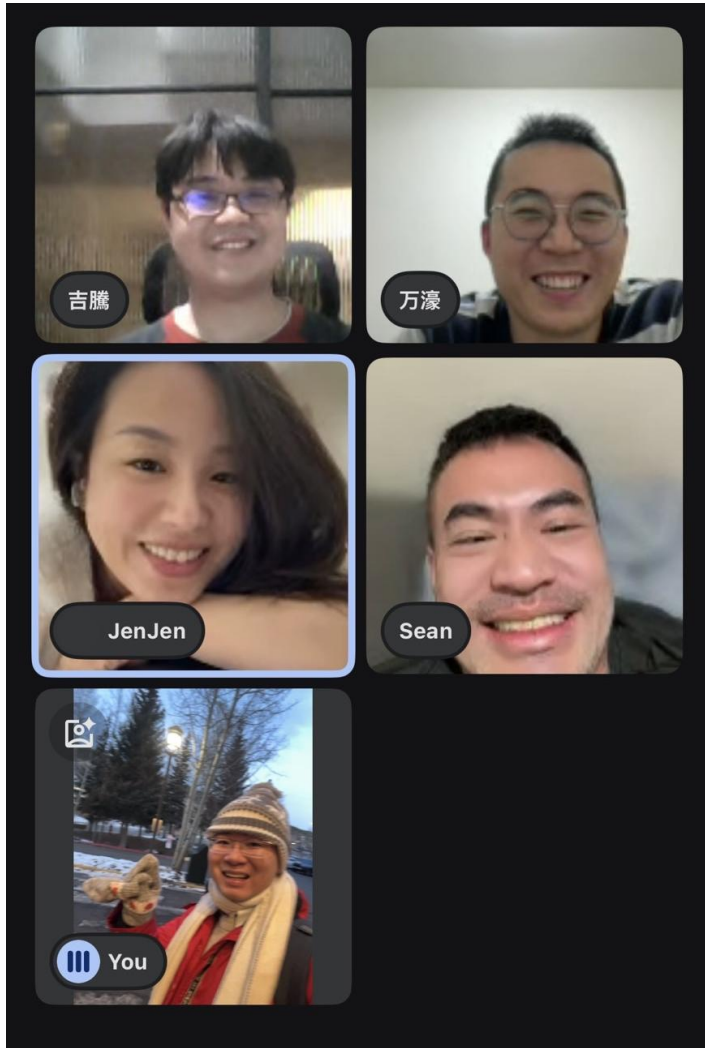


Dr. Johannes Roth

Sponsor : 米飛生醫



特別誌謝 (Special Thanks)



基隆長庚醫院 許智欽教授
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悅兒診所 黃柏翰醫師
林口長庚 林芷儀、蒙彥傑
PReS MSUS Groups
TCR年輕醫師工作小組
My wife and family

Reference

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- 楊舒媚、林家瑋、陳文翔 台灣復健醫誌 2022; 50(2): 135 – 144
- P. Collado et al. *Pediatr Rheumatol Online J*. 2018 Apr 10;16(1):23.
- Roth J, Jousse-Joulin S et al. *Arthritis Care Res (Hoboken)*. 2015 Jan;67(1):136-42.

Let's Do it !!!

Hands-on time

